

Here is the expanded, fully detailed Software Requirements Specification (SRS) containing all functional parameters, data schemas, role permissions, state transitions, API specs, and operational workflows integrated into the core scope.

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

DG Fuel Management, Maintenance, Vendor Workflow & Billing

Project: Flipkart Minutes - Dark Store Operations

Platform: IoSense Platform

Document Status: Final Scope Specification

1. Purpose & Core Scope

1.1 Purpose

This document defines the end-to-end software requirements for real-time Diesel Generator (DG) IoT monitoring, preventive and breakdown maintenance management, vendor refueling workflows, line-item vendor billing, and automated ERP/Fusion integration across Flipkart Minutes dark store facilities.

1.2 Scope Architecture

- **Core Base Scope:** Store and DG setup (primary, standby, and mobile DGs), multi-vendor mapping, automated and manual triggers (fuel-level based refilling, running-hours and breakdown maintenance), cross-stakeholder maintenance visibility, vendor refueling execution, itemized vendor billing (refill charges, diesel/running hour rates), and Oracle Fusion API integration.
- **Optional Scope:** Automated report scheduling and distribution engine.

2. System Entities, Setup & Master Data

2.1 Geographic & Store Hierarchy

- **Hierarchy:** Zone → City → Store / Dark Store Site.
- **Store Parameters:** Store ID, Store Name, Zone, City, Store Address, Primary SPOC Contact, assigned Store Users.

2.2 Multi-DG Setup & IoT Onboarding

Each store site supports one or more DGs operating under three distinct roles:

1. **Primary DG:** Main operational backup generator.
2. **Standby DG:** Secondary installed generator for extended power outages.
3. **Mobile DG:** Temporary mobile unit deployed for emergency capacity or maintenance swaps.

Real-Time Data Captured per DG via IoSense IoT Devices:

- **Fuel Level (%):** Real-time tank capacity percentage.
- **Estimated Fuel Volume (L):** Calculated current fuel volume in liters.
- **Cumulative Running Hours (Hrs):** Engine runtime meter readings.
- **Energy Generation (kWh):** Total power output produced.
- **Operating Status:** Live state (**Running**, **Idle**, **Off**, **Fault/Breakdown**).

3. Vendor Mapping, Scope Exposure & Roles

3.1 Multi-Vendor Mapping (Risk Mitigation)

To prevent operational single-point failures, a single store or DG can have multiple active vendors assigned simultaneously:

- **Primary Refueling Vendor:** Default vendor for automated low-fuel work orders.
- **Secondary / Standby Vendor:** Contingency vendor for manual or failover refueling.
- **AMC Maintenance Vendor:** Vendor responsible for preventive and corrective engine maintenance.
- **SPOC Management:** Vendors can register multiple single-point-of-contact (SPOC) profiles per site. If a vendor or SPOC becomes inactive, the system flags the DG as requiring mapping while retaining historical transaction logs.

3.2 Site Billing Type Mapping

Vendors must explicitly define and configure the **Billing Type** for each assigned store/DG before generating invoices:

- **Fixed Rate AMC**
- **Per Litre Fuel Rate**
- **Running Hours Rate**

3.3 Cross-Stakeholder Maintenance Exposure

The platform exposes real-time maintenance and breakdown data across all relevant operational stakeholders:

- **Shared Visibility:** Open breakdown tickets, preventive maintenance schedules, repair history, and engine diagnostic logs managed by third-party maintenance vendors are made visible to **DG AMC Vendors**, **Internal Maintenance Teams**,

Store Users, and Central Users.

3.4 Role-Based Access Control (RBAC) Matrix

Role	Hierarchy Access	Operational Permissions	Billing & Financial Permissions
Store User	Assigned Store(s)	Views live DG telemetry, receives alerts, triggers manual refill/breakdown work orders.	Views store refueling & maintenance history.
Central User	All Stores / Zones / Cities	Manages store master uploads, views global dashboards, overrides triggers, filters operational data.	Views & exports all vendor transactions and billing statuses.
Vendor User / SPOC	Assigned DGs / Sites	Acknowledges work orders, logs refilling (liters, dates, proof), logs maintenance resolution.	Selects billing periods, reviews eligible line items, submits invoices to Fusion.
Maintenance Team / Vendor	Assigned DGs	Receives breakdown alerts, logs preventive/corrective maintenance details, updates unit availability.	Views service order logs and maintenance billing claims.

Super Admin	Platform-wide	Configures global fuel/runtime thresholds, waiting timers, system settings, user roles, system overrides.	Full system audit trail access and configuration management.
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4. Work Order Triggers & Execution Workflows

4.1 Refilling Work Order Triggers

- **Automatic Trigger (Fuel Level Based):**
 - **Threshold Priority:** DG Level > Store Level > City Level > Zone Level > Global Level.
 - **Evaluation Rule:** When DG Fuel Level (%) drops below the configured limit and remains below it continuously for a set waiting timer (e.g., 15 minutes), the system automatically creates a unique **Refill Work Order ID** and assigns it to the primary active refueling vendor.
 - **De-duplication:** Prevents generating duplicate work orders for an ongoing low-fuel state.
- **Manual Trigger (On-Demand Refill):**
 - Store Users, Central Users, or Super Admins can manually generate a Refill Work Order at any time (e.g., severe weather warnings or grid outage notices).

4.2 Maintenance Work Order Triggers

- **Automatic Trigger (Running Hours Based):**
 - **Runtime Thresholds:** System tracks cumulative DG running hours via IoT. Upon reaching pre-set operational intervals (e.g., every 250 Hrs or 500 Hrs), the system automatically creates a **Preventive Maintenance Work Order** assigned to the maintenance vendor.
- **Manual Trigger (Breakdown / Corrective Maintenance):**
 - Store Users or Maintenance Teams can manually create a **Breakdown Work Order** upon physical failure, liquid leak, or starting failure.

4.3 Work Order State Machine

$\text{Created} \rightarrow \text{Assigned} \rightarrow \text{Acknowledged} \rightarrow \text{In Execution} \rightarrow \text{Submitted} \rightarrow \text{Completed}$

1. **Created:** Trigger condition met (auto or manual).
2. **Assigned:** System routes work order to the mapped primary/secondary vendor SPOC. Initial notification sent.

3. **Acknowledged:** Vendor SPOC confirms receipt via platform.
4. **In Execution:** Vendor performs refueling or maintenance at the dark store.
5. **Submitted:** Vendor enters operational data:
 - *Refill Data:* Liters added, date of refueling, optional proof upload.
 - *Maintenance Data:* Action taken, parts replaced, operational clearance.
6. **Completed:** Work order closed; data permanently locked to the transaction ID.

5. Core Vendor Billing & Line-Item Engine

5.1 Line-Item Separation & Rates

To prevent billing errors, every refueling transaction requires explicit line-item breakdowns:

1. **Refilling Charges (Line Item 1):** Vendor-specific service/handling fee per refueling event based on pre-aligned rate cards.
2. **Fuel / Consumption Charges (Line Item 2):** Separate rate calculation dependent on site billing type:
 - **Diesel Rate Item:** Fuel Liters Added $\times$ Agreed Rate per Liter.
 - **Running Hours Rate Item:** Operational Hours Run $\times$ Agreed Rate per Hour.

5.2 End-to-End Billing & Oracle Fusion Workflow

1. **Period Selection:** Vendor selects the billing cycle/date range.
2. **Method Selection:** Vendor selects calculation basis (Run Hours vs. Fuel Consumption).
3. **Transaction Rendering:** Platform pulls all unbilled, completed transactions matching criteria. Previously billed transactions are automatically excluded.
4. **Itemization Verification:** Platform presents line-item breakdowns (Refill Charges + Fuel/Hour Rates) per transaction.
5. **Invoice Entry & Submission:** Vendor enters total invoice number, attaches PDF copy, and submits.
6. **Transaction Lock & Fusion Posting:** The system permanently locks all included Work Order IDs against duplicate billing and transmits the structured payload to **Oracle Fusion via API**.

6. Data Structure & Technical Attributes

6.1 Key Data Schema

- **Store Master:** StoreID, StoreName, Zone, City, Address, EmailList.
- **DG Master:** DGID, StoreID, DGType (Primary/Standby/Mobile), Capacity_kVA, IoTDeviceID, FuelCapacity_L.
- **Vendor Mapping:** MappingID, DGID, VendorID, VendorType (Refuel/AMC), BillingType, IsActive.

- **Work Order Record:** `WorkOrderID`, `DGID`, `VendorID`, `SPOCID`, `TriggerType` (`Auto_Fuel` / `Manual_Fuel` / `Auto_Hrs` / `Manual_Breakdown`), `Status`, `CreationTimestamp`, `FuelAdded_L`, `RefillDate`, `ProofURL`.
- **Billing Record:** `InvoiceID`, `VendorID`, `BillingPeriod`, `WorkOrderList`, `TotalRefillCharges`, `TotalFuelCharges`, `InvoiceAmount`, `FusionStatus`, `SubmissionTimestamp`.

7. Dashboards, Audit Trail & System Logs

- **Live Operational Dashboard:** Real-time visibility into store fuel levels (%), DG running hours, active alerts, pending refilling work orders, and open maintenance/breakdown tickets.
- **Audit Trail:** Immutable logging of all system actions:
 - Automatic trigger conditions and timestamped evaluations.
 - Manual work order overrides and creator user IDs.
 - Vendor work order acknowledgements and data submissions.
 - Invoice submission history and Oracle Fusion API success/failure responses.
- **Data Export:** Filterable operational and financial export tools (CSV/Excel) supporting filters by Zone, City, Store, DG, Vendor, Date Range, and Status.